NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

A VERY LONG AND IMPRESSIVE THESIS TITLE WITH A FORCED LINE BREAK

SOME THOUGHTS ON THE LIFE, THE UNIVERSE, AND EVERYTHING ELSE

JOHN VERY LONGNAME DOE

Master/BSc in Name of Previous Degree

DOCTORATE IN AEROSPACE ENGINEERING

SPECIALIZATION IN SPECIALIZATION NAME

NOVA University Lisbon

September, 2000

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Dedicatory lorem ipsum.

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Acknowledgments are personal text and should be a free expression of the author.

However, without any intention of conditioning the form or content of this text, I would like to add that it usually starts with academic thanks (instructors, etc.); then institutional thanks (Research Center, Department, Faculty, University, FCT / MEC scholarships, etc.) and, finally, the personal ones (friends, family, etc.).

But I insist that there are no fixed rules for this text, and it must, above all, express what the author feels.

"You cannot teach a man anything; you can only help him discover it in himself."

— **Galileo**, Somewhere in a book or speach
(Astronomer, physicist and engineer)

Abstract

Resumo

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Acronyms

CBF Control Barrier Function *(pp. vii, 3–11, 14–16)*

CLF Control Lyapunov Function *(pp. 2–7, 11, 14–17)*

CLF-CBF Control Lyapunov Function - Control Barrier Function *(pp. vii, 2, 3, 5, 7–9, 11, 14, 15, 17)*

KKT Karush-Kuhn-Tucker *(pp. 7, 14, 16, 17)*

QP quadratic program *(pp. vii, 4–8, 11, 14, 15, 17)*

1

Introduction

2

Methodology

2.1 CLF-CBF

Considering a nonlinear control-affine system:

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} \tag{2.1}$$

with stet $\mathbf{x} \in \mathbb{R}^n$, control input $\mathbf{u} \in \mathbb{R}^m$, and locally Lipschitz continuos[1] functions $f : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^{n\times m}$. Given a locally Lipschitz continuos $k : \mathbb{R}^n \to \mathbb{R}^m$, it results in the closed-loop system:

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})k(\mathbf{x}) \tag{2.2}$$

For any initial condition $\mathbf{x}_0 \in \mathbb{R}^n$, there is a maximal time interval $\mathcal{I}(\mathbf{x}_0) = [0; t_{max}(\mathbf{x}_0)[$ and unique continuosly differentiable solution $\varphi : \mathcal{I}(\mathbf{x}_0) \to \mathbb{R}^n$ such that:

$$\begin{aligned} &\dot{\varphi}(t) = f(\varphi(t)) + g(\varphi(t))k(\varphi(t)) \quad \forall t \in \mathcal{I}(\mathbf{x}_0) \\ &\varphi(0) = \mathbf{x}_0 \end{aligned} \tag{2.3}$$

2.1.1 First-Order Formulation

2.1.1.1 Control Lyapunov Function (CLF)

Having the objective of global assymptotically stability of the system 2.2, is desired for the system converge to an equilibrium point $\bar{\mathbf{x}}$. A solution comes by designing a positive definite function $V : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ which converges to zero while the system converges to the respective equilibrium point, and for that, designing a control law that acts in order to achieve specifically the convergence of the function. Consequentially is pretended to the function $V(\mathbf{x})$ to be continuos and differentiable:

$$\dot{V}(\mathbf{x}, \mathbf{u}) = \frac{\partial V}{\partial x}\dot{x} = L_f V(\mathbf{x}) + L_G V(\mathbf{x})\mathbf{u}$$

where $L_f V$ and $L_G V$ are the Lie derivatives of V and $\mathbf{u}$ the control input vector resposible to manipulate the function, more precisely, its convergence rate.

[1] Given two metric spaces (X, d_X) and (Y, d_Y), where d_X and d_Y represent the X and Y metrics respectively, a function $f : X \to Y$ is Lipschitz continuos if there exist a constant $L \geq 0$ such that $\forall (x_1, x_2) \in X, \quad d_Y(f(x_1), f(x_2)) \leq L\, d_X(x_1, x_2)$

That beeing said, a continuosly differentiable, positive definite function $V : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ wich $\lim_{\mathbf{x}\to\bar{\mathbf{x}}} V(\mathbf{x}) = 0$ is a CLF if there exists a $\gamma \in \mathcal{K}_\infty^e$ [2] such that for all $\mathbf{x} \in \mathbb{R}^n \setminus \{\bar{\mathbf{x}}\}$:

$$\inf_{\mathbf{u}\in\mathbb{R}^m} [L_f V(\mathbf{x}) + L_G V(\mathbf{x})\mathbf{u}] \leq -\gamma(V(\mathbf{x})) \tag{2.4}$$

From CLF V and an adequate tuning of γ parameter results a control law followed by the controller K_{CLF}:

$$K_{CLF}(\mathbf{x}) = \{\mathbf{u} \in \mathbb{R}^m : L_f V(\mathbf{x}) + L_G V(\mathbf{x})\mathbf{u} \leq -\gamma(V(\mathbf{x}))\} \tag{2.5}$$

The stets generated by $K_{CLF}(\mathbf{x})$ $\forall x \in \mathbb{R}^n$ refer to the possible control input values necessary to the CLF $V(\mathbf{x})$ continuosly converge to zero while the closed-loop system 2.2 to the equilibrium point, but more slowly as is close to reach the objective.

2.1.1.2 Control Barrier Function (CBF)

Now having the objective of garantee safety to system 2.2, is aimed to keep the state inside a safe set C (foward invariant[3]) defined accordingly the problem needs. The notion of safety, i.e., the safe set C is defined as the 0-superlevel[4] set of a continuosly differentiable function $h : \mathbb{R}^n \to \mathbb{R}$:

$$\begin{aligned} C &= \{\mathbf{x} \in \mathbb{R}^n : h(\mathbf{x}) \geq 0\} \\ \partial C &= \{\mathbf{x} \in \mathbb{R}^n : h(\mathbf{x}) = 0\} \\ int(C) &= \{\mathbf{x} \in \mathbb{R}^n : h(\mathbf{x}) > 0\} \end{aligned} \tag{2.6}$$

In order to mantain $h(\mathbf{x})$ positive, there is a need to have a control law and but first to be able to interface with the function. Since $h(\mathbf{x})$ is differentiable:

$$\dot{h}(\mathbf{x}, \mathbf{u}) = \frac{\partial h}{\partial x}\dot{x} = L_f h(\mathbf{x}) + L_G h(\mathbf{x})\mathbf{u}$$

where $L_f h$ and $L_G h$ are the Lie derivatives of h and $\mathbf{u}$ the control input vector resposible to manipulate the function, more precisely, its time derivative.

So, a differentiable function $h : \mathbb{R}^n \to \mathbb{R}$ is a CBF if there exists a $\alpha \in \mathcal{K}_\infty^e$ such that for all $\mathbf{x} \in \mathbb{R}^n \setminus \{\bar{\mathbf{x}}\}$:

$$\sup_{\mathbf{u}\in\mathbb{R}^m} [L_f h(\mathbf{x}) + L_G h(\mathbf{x})\mathbf{u}] \geq -\alpha(h(\mathbf{x})) \tag{2.7}$$

Similar to the CLF, from CBF h and an adequate tuning of α parameter results a control law encapsulated by the controller K_{CBF}:

$$K_{CBF}(\mathbf{x}) = \{\mathbf{u} \in \mathbb{R}^m : L_f h(\mathbf{x}) + L_G h(\mathbf{x})\mathbf{u} \geq -\alpha(h(\mathbf{x}))\} \tag{2.8}$$

[2] If a continuos functions $\alpha : [-b, a] \to \mathbb{R}$ with $(a, b) > 0$, strictly increasing and $\alpha(0) = 0$ then $\alpha \in \mathcal{K}^e$ and if it adds $\lim_{r\to\infty} \alpha(r) = \infty$ then $\alpha \in \mathcal{K}_\infty^e$

[3] A set $\mathcal{S}$ is said foward invariant if the initial system state $x_0 \in \mathcal{S}$ and so $x(t) \in \mathcal{S}$ $\forall t \geq 0$

[4] For function $f : \mathbb{R}^n \to \mathbb{R}$, its superlevel set is given by $\mathcal{S}(f) = \{(x_1, ..., x_n) : f(x_1, ..., x_n) \geq c\}$, where having $c = 0$ corresponds to the 0-superlevel set of f

The controller $K_{CLF}(\mathbf{x})\ \ \forall x \in \mathbb{R}^n$ renders control input sets that keep the closed-loop system 2.2 safe with respect to the set C. The controller by defenition is able to evolve the closed-loop system more freely when $\dot{h}(\mathbf{x}) > 0$ and more cautiously as $h(\mathbf{x})$ shortens and $\dot{h}(\mathbf{x}) < 0$, if $h(\mathbf{x}) < 0$ it means is in the unsafe region, so $\dot{h}(\mathbf{x}) > 0$ implies converging continuosly to the safe set C.

2.1.1.3 Quadratic Program Formulation

Now having the objective of not only achieve safety but also assymptotically stability through the usage of CLF and CBF it renders a set of control inputs able to achieve the respective purposes.

Using the conditions imposed by CLF and CBF as constraints, is presented an optimization problem via quadratic program (QP) [5], given a locally Lipschitz continuos controller $k : \mathbb{R}^n \longrightarrow \mathbb{R}^m$:

$$\begin{aligned} k(\mathbf{x}) = \underset{\mathbf{u}\in\mathbb{R}^m}{\operatorname{argmin}}\ & \tfrac{1}{2}(\mathbf{u}^\top R\mathbf{u}) \\ st.\ & L_f V(\mathbf{x}) + L_G V(\mathbf{x})\mathbf{u} \leq -\gamma(V(\mathbf{x})) \\ & L_f h(\mathbf{x}) + L_G h(\mathbf{x})\mathbf{u} \geq -\alpha(h(\mathbf{x})) \end{aligned} \tag{2.9}$$

The sets of control inputs restricted to the optimization problem are given by $K_{CLF-CBF}(\mathbf{x}) = K_{CLF}(\mathbf{x}) \cap K_{CBF}(\mathbf{x})$ hopefully non-empty which can happen if the conditions are in conflict.

In case of conditions' conflict, in order to the closed loop system 2.2 controller have a solution from the QP, since safety is critical, using a slack variable δ allows to relax the stability condition imposed by the CLF and highly contribute to obtaining a solution while keeping system safe:

$$\begin{aligned} k(\mathbf{x}) = \underset{\mathbf{u}\in\mathbb{R}^m,\ \delta\in\mathbb{R}}{\operatorname{argmin}}\ & \tfrac{1}{2}(\mathbf{u}^\top R\mathbf{u} + p\delta^2) \\ st.\ & L_f V(\mathbf{x}) + L_G V(\mathbf{x})\mathbf{u} \leq -\gamma(V(\mathbf{x})) + \delta \\ & L_f h(\mathbf{x}) + L_G h(\mathbf{x})\mathbf{u} \geq -\alpha(h(\mathbf{x})) \end{aligned} \tag{2.10}$$

Having δ as an optimization variable not only allows to balance between the input values exceed usage and the level of relaxation but also to not have a fixed δ value which would result for all states not so faster convergence and to the neighborhood of the equilibrium point instead of the actual target point.

Notes:

- If it's not pretended to enforce stability or safety, just remove from the optimization problem the respective constraint responsible to it

[5] quadratic program consists of solving a mathematical optimization problem with a quadratic objective function

2.1.2 Backstepping (Second-Order Formulation)

2.1.2.1 Second-Order Systems

Usually control systems are governed by higher-order systems, such as some second-order cyber-physical systems controlling the position by applying some force to manipulate the speed that is felt in the position just instants after. Higher order systems' outputs variation is felt by an multiple integrative effect, by an integration of each state affecting the next state dynamic until reach the so wanted output. Considering a non-linear system in strict-feedback form:

$$\dot{\mathbf{x}} = f_0(\mathbf{x}) + g_0(\mathbf{x})\xi \tag{2.11a}$$

$$\dot{\xi} = f_1(\mathbf{x}, \xi) + g_1(\mathbf{x}, \xi)\mathbf{u} \tag{2.11b}$$

with $\mathbf{x} \in \mathbb{R}^n$, $\xi \in \mathbb{R}^p$, and $\mathbf{u} \in \mathbb{R}^m$, and functions $f_0 : \mathbb{R}^n \to \mathbb{R}^n$, $g_0 : \mathbb{R}^n \to \mathbb{R}^{n\times p}$, $f_1 : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^p$ and $g_1 : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^{p\times m}$ locally Lipschitz continuos on their respective domains.

Given a locally Lipschitz continuos feedback controller $\mathbf{k} : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^m$ the closed-loop system is:

$$\begin{aligned} \dot{\mathbf{x}} &= f_0(\mathbf{x}) + g_0(\mathbf{x})\xi \\ \dot{\xi} &= f_1(\mathbf{x}, \xi) + g_1(\mathbf{x}, \xi)\mathbf{k}(\mathbf{x}, \xi) \end{aligned} \tag{2.12}$$

for any initial states $(\mathbf{x}_0, \xi_0) \in \mathbb{R}^n \times \mathbb{R}^p$ exists a maximum time interval $I(\mathbf{x}_0, \xi_0) \subseteq \mathbb{R}_{\geq 0}$ and a unique solution $\varphi = (\varphi_{\mathbf{x}}, \varphi_\xi)$ satisfying 2.3 $\forall t \in I(\mathbf{x}_0, \xi_0)$.

2.1.2.2 Purposal Solution Obtaining

Since having a second order system 2.11 the access to ξ is not direct, we must "backstepp" in order to be able to control $\mathbf{u}$, affect ξ and consequentially control the wanted dynamic $\dot{\mathbf{x}}$. The strategy passes by obtaining the solution if it was a first order system 2.1 and if ξ could be directly controlled (i.e. as part of the first order system control input), so given a differentiable and locally Lipschitz continuos $k_0(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}^p$ (that encompasses the "controllable ξ"), the closed-loop system:

$$\dot{\mathbf{x}} = f_0(\mathbf{x}) + g_0(\mathbf{x})k_0(\mathbf{x}) \tag{2.13}$$

That beeing said, the first order solution could be obtained via QP, so supposing a CLF V_0, CBF h_0, a differentiable and locally Lipschitz continuos described just now $k_0(\mathbf{x})$ and $(\gamma_0, \alpha_0) \in \mathcal{K}^e_\infty \times \mathcal{K}^e_\infty$:

$$\begin{aligned} k_0(\mathbf{x}) = \underset{\mathbf{u}\in\mathbb{R}^p,\ \delta\in\mathbb{R}}{\operatorname{argmin}} \ & \tfrac{1}{2}(\mathbf{u}^\top R_0 \mathbf{u} + p_0\delta^2) \\ st.\ \ & L_f V_0(\mathbf{x}) + L_G V_0(\mathbf{x})\mathbf{u} \leq -\gamma_0(V_0(\mathbf{x})) + \delta \\ & L_f h_0(\mathbf{x}) + L_G h_0(\mathbf{x})\mathbf{u} \geq -\alpha_0(h_0(\mathbf{x})) \end{aligned} \tag{2.14}$$

The solution is equivalent to a value of ξ closest to an optimal value taken by it that would impose safety and contribute to stability not only to the first-order system 2.13 but also to the actual second-order system. Although like it was said is not possible to have direct access to ξ,

the backstepping technique is essentially the second-order trying to converge to the first order solution. Supposing there is $V_\xi(\mathbf{x}, \xi) : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}_{\geq 0}$, the new CLF $V(\mathbf{x}, \xi)$ and CBF $h(\mathbf{x}, \xi)$ are:

$$V_\xi(\mathbf{x}, \xi) = \frac{1}{2}\|\xi - k_0(\mathbf{x})\|^2 \tag{2.15a}$$

$$V(\mathbf{x}, \xi) = V_0(\mathbf{x}) + V_\xi(\mathbf{x}, \xi) \tag{2.15b}$$

$$h(\mathbf{x}, \xi) = h_0(\mathbf{x}) - V_\xi(\mathbf{x}, \xi) \tag{2.15c}$$

Finally, via QP, given a differentiable and locally Lipschitz continuos $k(\mathbf{x}, \xi) : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^m$:

$$\begin{aligned} k(\mathbf{x}, \xi) = \underset{\mathbf{u}\in\mathbb{R}^p,\, \delta\in\mathbb{R}}{\operatorname{argmin}} \ & \tfrac{1}{2}(\mathbf{u}^\top R\mathbf{u} + p\delta^2) \\ st. \ & L_f V(\mathbf{x}, \xi) + L_G V(\mathbf{x}, \xi)\mathbf{u} \leq -\gamma(V_0(\mathbf{x})) - \gamma'(V_\xi(\mathbf{x}, \xi)) + \delta \\ & L_f h(\mathbf{x}, \xi) + L_G h(\mathbf{x}, \xi)\mathbf{u} \geq -\alpha(h_0(\mathbf{x})) + \alpha'(V_\xi(\mathbf{x}, \xi)) \end{aligned} \tag{2.16}$$

It's expected that $\gamma << \gamma'$ contributing to a faster convergence of ξ and more similar behaviour of the system 2.12 relative to the first-order system 2.13 and mitigate the chances of the system 2.12 not be safe in relation to C_0.

Notes:

- If it's not pretended to enforce stability or safety, just remove from both optimization problem (2.14 and 2.16) the respective constraint responsible to it
- If it's a higher-order (than second) system, then, assuming it's a third-order system, the solution obtained in 2.16 is the "new k_0" and the third-order solution tries to converge to the second-order one (while this one is converging to the first-order solution)

2.1.3 Closed Form

The objective of this thesis is to purpose a set of low computational effort techniques suitable to spacecrafts. Aiming a faster computation of the optimization problems solutions, any of the previous QPs (2.9, 2.10, 2.14, 2.16) have a closed-form[6] solution which allows to calculate directly the respective solution instead of depending of a numerical solver. Given the QP optimization problem and a controller $k(\mathbf{x}, \xi) : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^m$ (if it is a second-order system):

$$\begin{aligned} k(\mathbf{x}, \xi) = \underset{\mathbf{u}\in\mathbb{R}^m,\, \delta\in\mathbb{R}}{\operatorname{argmin}} \ & \tfrac{1}{2}(\mathbf{u}^\top R\mathbf{u} + p\delta^2) \\ st. \ & \underbrace{L_f V(\mathbf{x}, \xi) + \gamma(V_0(\mathbf{x})) + \gamma'(V_\xi(\mathbf{x}, \xi))}_{F_V(\mathbf{x},\xi)} + L_G V(\mathbf{x}, \xi)\mathbf{u} - \delta \leq 0 \\ & \underbrace{L_f h(\mathbf{x}, \xi) + \alpha(h_0(\mathbf{x})) - \alpha'(V_\xi(\mathbf{x}, \xi))}_{F_h(\mathbf{x},\xi)} + L_G h(\mathbf{x}, \xi)\mathbf{u} \geq 0 \end{aligned} \tag{2.17}$$

[6] An closed-form expression is expressed by a limit number and well defined operations (elementary functions)

With a CLF V, CBF h, $\mathbf{x} \in \mathbb{R}^n$, the first-order system CLF V_0(= V like in 2.9, 2.10 and 2.14) , and if it is a second-order system with $\xi \in \mathbb{R}^p$ (equivalent to the QP seen while doing Backstepping 2.16).

Using [2] as a reference, assuming equal control input wheights relative to each other ($R = \mathbf{I}_{m\times m}$) and just one CLF and CBF each, according to the Karush-Kuhn-Tucker (KKT) conditions[7] (to see the full deduction A.1):

$$\mathbf{k}(\mathbf{x}, \xi) = \begin{cases} \mathbf{k}_1(\mathbf{x}, \xi), & (\mathbf{x}, \xi) \in \mathcal{S}_1 \\ \mathbf{k}_2(\mathbf{x}, \xi), & (\mathbf{x}, \xi) \in \mathcal{S}_2 \\ \mathbf{k}_3(\mathbf{x}, \xi), & (\mathbf{x}, \xi) \in \mathcal{S}_3 \\ \mathbf{0}_{m\times 1} \quad , & (\mathbf{x}, \xi) \in \mathcal{S}_4 \end{cases} \tag{2.18}$$

Where each controller $\mathbf{k}_i$, for $i = 1, ..., 4$, corresponds to the closed-form equation depending which constraints (CLF or CBF) are active what can be inferred by $\mathcal{S}_i$, for $i = 1, ..., 4$. Via KKT conditions the respectives controllers and domains are obtained due to the KKT multipliers.

Auxiliary Variables:

$$\begin{aligned} L(\mathbf{x}, \xi) &= L_G V(\mathbf{x}, \xi)\, L_G h(\mathbf{x}, \xi)^\top \\ \Delta &= L(\mathbf{x}, \xi)^2 (p^{-1} + \|L_G h(\mathbf{x}, \xi)\|^2)\, \|L_G h(\mathbf{x}, \xi)\|^2 \end{aligned}$$

KKT multipliers:

$$\begin{aligned} \lambda_1(\mathbf{x}, \xi) &= \Delta^{-1}(L(\mathbf{x}, \xi)\, F_h(\mathbf{x}, \xi) \; - \|L_G h(\mathbf{x}, \xi)\|^2\, F_V(\mathbf{x}, \xi)) \\ \lambda_2(\mathbf{x}, \xi) &= \Delta^{-1}(L(\mathbf{x}, \xi) F_V(\mathbf{x}, \xi) \; - (p^{-1} + \|L_G h(\mathbf{x}, \xi)\|^2)\, F_h(\mathbf{x}, \xi)) \end{aligned} \tag{2.19}$$

Controller $\mathbf{k}_i$ closed-form equation:

$$\begin{aligned} \mathbf{k}_1(\mathbf{x}, \xi) &= -(p^{-1} + \|L_G V(\mathbf{x}, \xi)\|^2)^{-1}\, F_V(\mathbf{x}, \xi)\, L_G V(\mathbf{x}, \xi)^\top \\ \mathbf{k}_2(\mathbf{x}, \xi) &= -\|L_G h(\mathbf{x}, \xi)\|^{-2}\, F_h(\mathbf{x}, \xi)\, L_G h(\mathbf{x}, \xi)^\top \\ \mathbf{k}_3(\mathbf{x}, \xi) &= -\lambda_1(\mathbf{x}, \xi) L_G V(\mathbf{x}, \xi)^\top + \lambda_2(\mathbf{x}, \xi) L_G h(\mathbf{x}, \xi)^\top \end{aligned} \tag{2.20}$$

Constraint Activation Subdomains $\mathcal{S}_i$ (based on the dual feasibilitie condition [7]):

If CLF is active:

$$\mathcal{S}_1(\mathbf{x}, \xi) = \{(\mathbf{x}, \xi) \in \mathbb{R}^n \times \mathbb{R}^p : \lambda_1 \geq 0 \cap \lambda_2 < 0\}$$

If CBF is active:

$$\mathcal{S}_2(\mathbf{x}, \xi) = \{(\mathbf{x}, \xi) \in \mathbb{R}^n \times \mathbb{R}^p : \lambda_1 < 0 \cap \lambda_2 \geq 0\} \tag{2.21}$$

[7] KKT Conditions Wikipedia (Click here to be Redirected)

If CLF and CBF is active:

$$\mathcal{S}_3(\mathbf{x}, \xi) = \{(\mathbf{x}, \xi) \in \mathbb{R}^n \times \mathbb{R}^p : \lambda_1 \geq 0 \cap \lambda_2 \geq 0\}$$

If Any is active:

$$\mathcal{S}_4(\mathbf{x}, \xi) = \{(\mathbf{x}, \xi) \in \mathbb{R}^n \times \mathbb{R}^p : \lambda_1 < 0 \cap \lambda_2 < 0\}$$

2.1.4 Unifying CBF

The previous controllers synthesized control inputs based on just one CBF, i.e., those controllers garantee safety w.r.t. a single safe set C, but given diffrent scenarios, it could exist more complex safety specifications and the need to impose more than one safety contraints simultaneously such as CBF $h_i(\mathbf{x})$, for $i \in \mathcal{I} = \{1, ...N\}$:

$$C_i = \{\mathbf{x} \in \mathbb{R}_n : h_i(\mathbf{x}) \geq 0\} \tag{2.22}$$

$$C = \bigcap_{i \in \mathcal{I}} C_i \tag{2.23}$$

But like it was said before regarding having more than one constraint could leave to conflict problems, empty safe sets ($C = \emptyset$) and an inexistent set of acceptable control inputs ($\mathbf{k}(\mathbf{x}, \xi) = \emptyset$). Using [3] as guidance, in order to avoid conflicts, merges all CBF into a single one that is able to capture all those safety notions and allow for even more complex safety specifications and to the resultant constraint fit the already obtained CLF-CBF closed-form solution 2.18.

2.1.4.1 First-Order System

The more complex safety specifications are defined via boolean logical operations between 0-superlevel sets of the diffrent CBF. Those operations are tipically seen in constrained optimization problems, like in QP of the form:

$$\begin{aligned} k(\mathbf{x}) = \underset{\mathbf{u} \in \mathbb{R}^p,\, \delta \in \mathbb{R}}{\text{argmin}} \quad & \tfrac{1}{2}(\mathbf{u}^\top R \mathbf{u} + p\delta^2) \\ st. \quad & L_f V(\mathbf{x}) + L_G V(\mathbf{x})\mathbf{u} \leq -\gamma(V(\mathbf{x})) + \delta \\ & L_f h_i(\mathbf{x}) + L_G h_i(\mathbf{x})\mathbf{u} \geq -\alpha_i(h_i(\mathbf{x})) \quad \forall i \in \mathcal{I} \end{aligned} \tag{2.24}$$

Where the resultant safe set 2.23 of the system refers to the intersection of all safe sets 2.22. More than be propitious to be not solvable, the formulation itself of the optimization problem 2.24 is limitated by no other logical operation than AND.

Among the the logical operations that could be applied to CBFs, there are:

Identity (Class-$\mathcal{K}^e$):

$$C_i = \{(\mathbf{x}, \alpha_i) \in \mathbb{R}^n \times \mathcal{K}^e : \alpha_i(h_i(\mathbf{x}) \geq 0)\} = \tag{2.25a}$$

$$= \{\mathbf{x} \in \mathbb{R}^n : h_i(\mathbf{x}) \geq 0\} \tag{2.25b}$$

Complement Set (Negation):

$$\bar{C}_i = \{\mathbf{x} \in \mathbb{R}^n : -h_i(\mathbf{x}) \geq 0\} \tag{2.26}$$

Intersection of Sets (AND):

$$\begin{aligned} \bigcap_{i \in \mathcal{I}} C_i &= \{\mathbf{x} \in \mathbb{R}^n : h_i(\mathbf{x}) \geq 0 \ \ \forall i \in \mathcal{I}\} \\ &= \{\mathbf{x} \in \mathbb{R}^n : \min_{i \in \mathcal{I}}(h_i(\mathbf{x})) \geq 0\} \end{aligned} \tag{2.27}$$

Union of Sets (OR):

$$\begin{aligned} \bigcup_{i \in \mathcal{I}} C_i &= \{\mathbf{x} \in \mathbb{R}^n : \exists i \in \mathcal{I} \ \ st. \ \ h_i(\mathbf{x}) \geq 0\} \\ &= \{\mathbf{x} \in \mathbb{R}^n : \max_{i \in \mathcal{I}}(h_i(\mathbf{x})) \geq 0\} \end{aligned} \tag{2.28}$$

The logical operations identity, complement, intersection and union are represented by respectively, class-$\mathcal{K}^e$ functions, negation, min and max operations. Although the class-$\mathcal{K}^e$ ($\alpha_i(h_i(\mathbf{x}))$) and negation ($-h_i(\mathbf{x})$) functions which are continuosly differentiable and valid as CBF, contrary to the AND and OR functions, $\min_{i \in \mathcal{I}}(h_i(\mathbf{x}))$ and $\max_{i \in \mathcal{I}}(h_i(\mathbf{x}))$ that aren't continuosly differentiable and therefore not CBF candidates. The max and min functions capacitate combining safe sets defined by multiple CBF (accordingly to the operation) and to obtain a new safety specification as is expected to see while merging multiple CBF into a single one. So, in order to viablize the AND and OR operation and achieve the merge of CBFs, the min and max functions can be replaced via smooth-aproximation, keeping their properties with just a relative small degradation of accuracy, by continuosly differentiable **log-sum-exp functions**:

Intersection of Sets \AND \ Minimum

$$\begin{aligned} h(\mathbf{x}) &= \frac{-1}{k} ln(\sum_{i \in \mathcal{I}} e^{-k\ h_i(\mathbf{x})}) \\ &\approx \min_{i \in \mathcal{I}}(h_i(\mathbf{x})) \end{aligned} \tag{2.29}$$

Beeing $k \in \mathbb{R}_{>0}$ as smoothing parameter (if $k < 0$ then is equivalent to the smooth-aproximation of $\max_{i \in \mathcal{I}}(h_i(\mathbf{x}))$).

$$\dot{h}(\mathbf{x}, \mathbf{u}) = \underbrace{\sum_{i \in \mathcal{I}} [\lambda_i \ L_f h_i(\mathbf{x})]}_{Lfh(\mathbf{x})} + \underbrace{(\sum_{i \in \mathcal{I}} [\lambda_i \ L_G h_i(\mathbf{x})])}_{LGh(\mathbf{x})}\mathbf{u} \tag{2.30a}$$

$$\lambda_i(\mathbf{x}) = e^{-k(h_i(\mathbf{x}) - h(\mathbf{x}))} \tag{2.30b}$$

With $\sum_{i \in \mathcal{I}} \lambda_i(\mathbf{x}) = 1$

Encountering an under-aproximation of h relative to the min function, given the safe set C defined based on 2.6 it verifies:

$$\min_{i \in \mathcal{I}}(h_i(\mathbf{x})) \geq h(\mathbf{x}) \quad \forall \mathbf{x} \in \mathbb{R}^n \tag{2.31a}$$

$$C \subseteq \bigcap_{i \in \mathcal{I}} C_i \tag{2.31b}$$

But, $\lim_{k\to\infty} h(\mathbf{x}) = \min_{i\in\mathcal{I}}(h_i(\mathbf{x}))$ and $\lim_{k\to\infty} C = \bigcap_{i\in\mathcal{I}} C_i$

On par with that, the new CBF $h(\mathbf{x})$ 2.29 could restrain too much the closed-loop system in comparison to theoretical CBF $\lim_{k\to\infty} h(\mathbf{x})$. In order to relax the safe set C and potentially contain $\bigcap_{i\in\mathcal{I}} C_i$:

$$h(\mathbf{x}) = \frac{-1}{k} ln(\sum_{i\in\mathcal{I}} e^{-k\ h_i(\mathbf{x})}) + \frac{b}{k} \tag{2.32}$$

Beeing b a parameter tuned according with the problem needs, knowing that if $b = ln(N)$:

$$\min_{i\in\mathcal{I}}(h_i(\mathbf{x})) \leq h(\mathbf{x}) \tag{2.33a}$$

$$C \supseteq \bigcap_{i\in\mathcal{I}} C_i \tag{2.33b}$$

Making the closed-loop system less constrained but not as safe.

Union of Sets \ OR \ Maximum

$$\begin{aligned} h(\mathbf{x}) &= \frac{1}{k} ln(\sum_{i\in\mathcal{I}} e^{k\ h_i(\mathbf{x})}) \\ &\approx \max_{i\in\mathcal{I}}(h_i(\mathbf{x})) \end{aligned} \tag{2.34}$$

Beeing $k \in \mathbb{R}_{>0}$ as smoothing parameter. The derivative of $h(\mathbf{x})$ is equal to 2.30a but:

$$\lambda_i(\mathbf{x}) = e^{k(h_i(\mathbf{x}) - h(\mathbf{x}))} \tag{2.35}$$

With also $\sum_{i\in\mathcal{I}} \lambda_i(\mathbf{x}) = 1$.

Due to an over-aproximation of h relative to the max function, given the safe set C defined based on 2.6 it verifies:

$$\max_{i\in\mathcal{I}}(h_i(\mathbf{x})) \leq h(\mathbf{x}) \quad \forall \mathbf{x} \in \mathbb{R}^n \tag{2.36a}$$

$$C \supseteq \bigcup_{i\in\mathcal{I}} C_i \tag{2.36b}$$

But, $\lim_{k\to\infty} h(\mathbf{x}) = \max_{i\in\mathcal{I}}(h_i(\mathbf{x}))$ and $\lim_{k\to\infty} C = \bigcup_{i\in\mathcal{I}} C_i$

Now showing a different point of view, safety is a critical factor, so in order to tight the safe set C and potentially be contained within $\bigcup_{i\in\mathcal{I}} C_i$:

$$h(\mathbf{x}) = \frac{1}{k} ln(\sum_{i\in\mathcal{I}} e^{k\ h_i(\mathbf{x})}) - \frac{b}{k} \tag{2.37}$$

Beeing b a tunable parameter. Taking $b = ln(N)$:

$$\max_{i\in\mathcal{I}}(h_i(\mathbf{x})) \geq h(\mathbf{x}) \tag{2.38a}$$

$$C \subseteq \bigcup_{i\in\mathcal{I}} C_i \tag{2.38b}$$

Therefore the higher b, the safer the system 2.13 but possibly too constrained by the new CBF $h(\mathbf{x})$ 2.37.

2.1.4.2 Second-Order System

Following the explanations given in the previous subsubsection 2.1.4.1, now applying for a second-order system 2.11, more precisely, for a Backstepping 2.1.2 situation.

Given a CLF $V_0(\mathbf{x})$: and multiple CBF $h0_i(\mathbf{x}) \quad \forall i \in \mathcal{I}$ combined into a CBF $h_0(\mathbf{x})$ via 2.32 or 2.37 (according it is an AND or OR operation), through the QP 2.14 (or its closed-form solution 2.18) is obtained a differentiable and locally Lipschitz continuos theoretical controller $k_0(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$.

In order to the second-order system 2.12 dynamic converge to $k_0(\mathbf{x})$ imposed dynamic, given multiple CBF $h_i(\mathbf{x}) \quad \forall i \in \mathcal{I}$:

$$V_\xi(\mathbf{x},\xi) = \frac{1}{2}\|\xi - k_0(\mathbf{x})\|^2 \tag{2.39a}$$

$$V(\mathbf{x},\xi) = V_0(\mathbf{x}) + V_\xi(\mathbf{x},\xi) \tag{2.39b}$$

$$h_i(\mathbf{x},\xi) = h0_i(\mathbf{x}) - V_\xi(\mathbf{x},\xi) \quad \forall i \in \mathcal{I} \tag{2.39c}$$

The resultant CBF $h_i(\mathbf{x},\xi) \quad \forall i \in \mathcal{I}$, one more time can be merged via 2.32 or 2.37 (according it is an AND or OR operation) into a single CBF $h(\mathbf{x},\xi)$. However there is another way in this case to have the new single CBF $h(\mathbf{x},\xi)$, so, considering the second-order system 2.11 and $lo \in \{\max, \min\}$:

$$\begin{aligned} \underset{i\in\mathcal{I}}{lo}(h0_i(\mathbf{x}) - V_\xi(\mathbf{x},\xi)) &= \underset{i\in\mathcal{I}}{lo}(h0_i(\mathbf{x})) - V_\xi(\mathbf{x},\xi) \approx \\ &\approx h_0(\mathbf{x}) - V_\xi(\mathbf{x},\xi) = \\ &= h(\mathbf{x},\xi) \end{aligned} \tag{2.40}$$

$$\dot{h}(\mathbf{x},\xi) = \underbrace{L_f h_0(\mathbf{x}) + L_G h_0(\mathbf{x})\xi - L_f V_\xi(\mathbf{x},\xi)}_{L_f h(\mathbf{x},\xi)} + \underbrace{(-L_G V_\xi(\mathbf{x},\xi))}_{L_G h(\mathbf{x},\xi)}\mathbf{u} \tag{2.41}$$

Having CBF h, the second-order solution, a differentiable and locally Lipschitz continuos controller $k(\mathbf{x},\xi) : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ is obtained using the QP 2.16 (or its closed-form solution 2.18).

Notes:

- There are other possible variations of a second-order system that invalidates 2.41 like the unicycle model that will be shown next chapter.

3

Results

Bibliography

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A

Deductions

A.1 Closed-Form QP given CLF-CBF Constraints

Following the QP 2.17, suppose there is only one CLF, given the differentiable quadratic objective function $f : \mathbb{R}^{m+1} \to \mathbb{R}$:

$$f(\mathbf{u}, \delta) = \frac{1}{2}(\mathbf{u}^\top R\mathbf{u} + p\delta^2) \tag{A.1}$$

where $\mathbf{u} \in \mathbb{R}^m$, $\delta \in \mathbb{R}$, $R \in \mathbb{R}^{m\times m}$ and $p \in \mathbb{R}$.

And the optimization problem constrained just by a CLF V and a CBF h presented in the respective differentiable functions $g_i(\mathbf{x}, \mathbf{u}, \delta) : \mathbb{R}^{n+m+1} \to \mathbb{R}, \quad \forall i \in \{1, 2\}$:

$$\begin{aligned} &\underbrace{F_V(\mathbf{x}) + L_G V(\mathbf{x})\mathbf{u} - \delta}_{g_1(\mathbf{x},\mathbf{u},\delta)} \leq 0 \\ &\underbrace{F_h(\mathbf{x}) + L_G h(\mathbf{x})\mathbf{u}}_{-g_2(\mathbf{x},\mathbf{u},\delta)} \geq 0 \end{aligned} \tag{A.2}$$

whith $\mathbf{x} \in \mathbb{R}^n$ (corresponding not to an optimzation variable but as a constant, as a current state of the system **??**).

The constrained optimization problem can form the Lagrangian function:

$$\mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda) = f(\mathbf{u}, \delta) + \lambda^\top g(\mathbf{x}, \mathbf{u}, \delta) \tag{A.3}$$

where $\lambda \in \mathbb{R}^2$ is called the KKT multiplier.

Aiming to hold the stationarity condition and find the optimal (minimal) solution:

$$\nabla(\mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda)) = 0 \tag{A.4}$$

Supposing $g(\mathbf{x}, \mathbf{u}, \delta) = \left[\begin{smallmatrix} g_1(\mathbf{x},\mathbf{u},\delta) \\ g_2(\mathbf{x},\mathbf{u},\delta) \end{smallmatrix}\right] = F(\mathbf{x}) + L_G(\mathbf{x})\mathbf{u} + \left[\begin{smallmatrix} -\delta \\ 0 \end{smallmatrix}\right]$ with $L_G(\mathbf{x}) = \left[\begin{smallmatrix} L_G V(\mathbf{x}) \\ -L_G h(\mathbf{x}) \end{smallmatrix}\right]$ and $F(\mathbf{x}) = \left[\begin{smallmatrix} F_V(\mathbf{x}) \\ -F_h(\mathbf{x}) \end{smallmatrix}\right]$:

$$\begin{aligned} \frac{\partial \mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda)}{\partial \mathbf{u}} = 0 &\iff \mathbf{u}^\top R + \lambda^\top L_G = 0 \\ &\iff R^\top \mathbf{u} + L_G^\top \lambda = 0 \\ &\iff \mathbf{u} = -(R^\top)^{-1} L_G^\top \lambda = -L_G^\top \lambda \quad \text{(Considering } R = \mathbf{I}_{m\times m} \\ &\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{i.e, assuming input consume equally)} \end{aligned} \tag{A.5}$$

$$\frac{\partial \mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda)}{\partial \delta} = 0 \iff p\delta + \lambda^\top \begin{bmatrix} -1 \\ 0 \end{bmatrix} = 0$$
$$\iff \delta = p^{-1} \begin{bmatrix} 1 & 0 \end{bmatrix} \lambda \tag{A.6}$$

$$\begin{aligned}
\frac{\partial \mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda)}{\partial \lambda} = 0 &\iff F(\mathbf{x}) + L_G(\mathbf{x})\mathbf{u} + \begin{bmatrix} -\delta \\ 0 \end{bmatrix} = 0 \\
&\iff F(\mathbf{x}) - \begin{bmatrix} \delta \\ 0 \end{bmatrix} - L_G(\mathbf{x})L_G^\top(\mathbf{x})\lambda = 0 \\
&\iff F(\mathbf{x}) - \begin{bmatrix} p^{-1} \begin{bmatrix} 1 & 0 \end{bmatrix} \lambda \\ 0 \end{bmatrix} - L_G(\mathbf{x})L_G^\top(\mathbf{x})\lambda = 0 \\
&\iff F(\mathbf{x}) - \left(\begin{bmatrix} p^{-1} & 0 \\ 0 & 0 \end{bmatrix} + L_G(\mathbf{x})L_G(\mathbf{x})^\top(\mathbf{x}) \right) \lambda = 0 \\
&\iff \lambda = \left(\begin{bmatrix} p^{-1} & 0 \\ 0 & 0 \end{bmatrix} + L_G(\mathbf{x})L_G(\mathbf{x})^\top(\mathbf{x}) \right)^{-1} F(\mathbf{x})
\end{aligned} \tag{A.7}$$

$$\left(\begin{bmatrix} p^{-1} & 0 \\ 0 & 0 \end{bmatrix} + L_G(\mathbf{x})L_G(\mathbf{x})^\top(\mathbf{x}) \right)^{-1} = \begin{bmatrix} p^{-1} + L_G V(\mathbf{x}) L_G V^\top(\mathbf{x}) & -L_G V(\mathbf{x}) L_G h^\top(\mathbf{x}) \\ -L_G h(\mathbf{x}) L_G V^\top(\mathbf{x}) & L_G h(\mathbf{x}) L_G h^\top(\mathbf{x}) \end{bmatrix}^{-1}$$
$$= \frac{adj(.)}{det(.)} = \frac{adj(.)}{\Delta} \tag{A.8}$$

$$adj(.) = \begin{bmatrix} L_G h(\mathbf{x}) L_G h^\top(\mathbf{x}) & L_G V(\mathbf{x}) L_G h^\top(\mathbf{x}) \\ L_G h(\mathbf{x}) L_G V^\top(\mathbf{x}) & p^{-1} + L_G V(\mathbf{x}) L_G V^\top(\mathbf{x}) \end{bmatrix} \tag{A.9}$$

$$\begin{aligned}
\Delta &= L_G h(\mathbf{x}) L_G V^\top(\mathbf{x}) L_G V(\mathbf{x}) L_G h^\top(\mathbf{x}) + (p^{-1} + L_G V(\mathbf{x}) L_G V^\top(\mathbf{x}))(L_G h(\mathbf{x}) L_G h^\top(\mathbf{x})) \\
&= (L_G h(\mathbf{x}) L_G V^\top(\mathbf{x}))^2 + (p^{-1} + \|L_G V(\mathbf{x})\|^2)\|L_G h(\mathbf{x})\|^2 \qquad \text{(Since there are just one CLF and CBF)}
\end{aligned} \tag{A.10}$$

$$\begin{aligned}
\lambda &= \Delta^{-1} adj(.) F(\mathbf{x}) \\
&= \Delta^{-1} \begin{bmatrix} -L_G V(\mathbf{x}) L_G h(\mathbf{x})^\top F_h + \|L_G h(\mathbf{x})\|^2 F_V \\ \\ L_G h(\mathbf{x}) L_G V(\mathbf{x})^\top F_V - (p^{-1} + \|L_G V(\mathbf{x})\|^2) F_h \end{bmatrix} \\
&= \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}
\end{aligned} \tag{A.11}$$

$$\begin{aligned}
\mathbf{u} &= -L_G^\top \lambda \\
&= -L_G V(\mathbf{x}) \lambda_1 + L_G h(\mathbf{x}) \lambda_2
\end{aligned} \tag{A.12}$$

According with the dual feasibility condition the respective constraint is active if $\lambda_i \geq 0$, for $i = \{1, 2\}$, so given the activation subdomain $\mathcal{S}_j$, for $j = \{1, 2, 3, 4\}$:

If any is active then:

$$\mathcal{S}_4(\mathbf{x}) = \{(\mathbf{x}) \in \mathbb{R}^n : \lambda_1 < 0 \cap \lambda_2 < 0\} \tag{A.13}$$

$$k_4 = \mathbf{0} \tag{A.14}$$

If both, the CLF and CBF is active then:

$$\mathcal{S}_3(\mathbf{x}) = \{(\mathbf{x}) \in \mathbb{R}^n : \lambda_1 \geq 0 \cap \lambda_2 \geq 0\} \tag{A.15}$$

$$k_3 = -L_G V(\mathbf{x})\lambda_1 + L_G h(\mathbf{x})\lambda_2 \tag{A.16}$$

If just the CBF is active:

$$\mathcal{S}_2(\mathbf{x}) = \{(\mathbf{x}) \in \mathbb{R}^n : \lambda_1 \geq 0 \cap \lambda_2 < 0\} \tag{A.17}$$

Or just the CLF is active:

$$\mathcal{S}_1(\mathbf{x}) = \{(\mathbf{x}) \in \mathbb{R}^n : \lambda_1 < 0 \cap \lambda_2 \geq 0\} \tag{A.18}$$

The equation A.12 is not suitable and doesn't support the KKT conditions and therefore there is a need to obtain the controller $k_j(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$, for $j = \{3, 4\}$, by removing from the optimization problem the respective inactive constraint.

Considering just the **CBF active**, the new Lagrangian function:

$$\mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_2) = f(\mathbf{u}, \delta) + \lambda_2^\top g_2(\mathbf{x}, \mathbf{u}, \delta) \tag{A.19}$$

One more time, keeping with the stationarity condition:

$$\nabla(\mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_2)) = 0 \tag{A.20}$$

$$\begin{aligned} \frac{\partial \mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_2)}{\partial \mathbf{u}} = 0 &\iff \mathbf{u}^\top R - \lambda_2^\top L_G h(\mathbf{x}) = 0 \\ &\iff R^\top \mathbf{u} - L_G h^\top(\mathbf{x})\lambda_2 = 0 \\ &\iff \mathbf{u} = (R^\top)^{-1} L_G h^\top(\mathbf{x})\lambda_2 = L_G h^\top \lambda_2 \quad (R = \mathbf{I}_{m\times m}) \end{aligned} \tag{A.21}$$

$$\begin{aligned} \frac{\partial \mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_2)}{\partial \delta} = 0 &\iff p\delta = 0 \\ &\iff \delta = 0 \end{aligned} \tag{A.22}$$

$$\begin{aligned}\frac{\partial \mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_2)}{\partial \lambda_2} = 0 &\iff -F_h(\mathbf{x}) - L_G h(\mathbf{x})\mathbf{u} = 0 \\ &\iff F_h(\mathbf{x}) + L_G h(\mathbf{x}) L_G h^\top(\mathbf{x})\lambda_2 = 0 \\ &\iff \lambda_2 = -(L_G h(\mathbf{x}) L_G h^\top(\mathbf{x}))^{-1} F_h(\mathbf{x}) \\ &\iff \lambda_2 = -\|L_G h(\mathbf{x})\|^{-2} F_h(\mathbf{x})\end{aligned} \tag{A.23}$$

Combinig the input equation A.21 and the KKT multiplier formula A.23:

$$\begin{aligned}k_2(\mathbf{x}) &= L_G h^\top(\mathbf{x})\lambda_2 \\ &= -L_G h^\top(\mathbf{x})\|L_G h(\mathbf{x})\|^{-2} F_h(\mathbf{x})\end{aligned} \tag{A.24}$$

Finally just with the **CLF active** :

$$\mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_1) = f(\mathbf{u}, \delta) + \lambda_1^\top g_1(\mathbf{x}, \mathbf{u}, \delta) \tag{A.25}$$

According with the stationarity condition:

$$\nabla(\mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_1)) = 0 \tag{A.26}$$

$$\begin{aligned}\frac{\partial \mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_1)}{\partial \mathbf{u}} = 0 &\iff \mathbf{u}^\top R + \lambda_1^\top L_G h(\mathbf{x}) = 0 \\ &\iff \mathbf{u} = -L_G V^\top \lambda_1 \quad (R = \mathbf{I}_{m\times m})\end{aligned} \tag{A.27}$$

$$\begin{aligned}\frac{\partial \mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_1)}{\partial \delta} = 0 &\iff p\delta - \lambda_1 = 0 \\ &\iff \delta = \frac{\lambda_1}{p}\end{aligned} \tag{A.28}$$

$$\begin{aligned}\frac{\partial \mathcal{L}(\mathbf{x}, \mathbf{u}, \delta, \lambda_1)}{\partial \lambda_1} = 0 &\iff F_V(\mathbf{x}) + L_G V(\mathbf{x})\mathbf{u} - \delta = 0 \\ &\iff F_V(\mathbf{x}) - L_G V(\mathbf{x}) L_G h^\top(\mathbf{x})\lambda_1 - \delta = 0 \\ &\iff F_V(\mathbf{x}) - L_G V(\mathbf{x}) L_G h^\top(\mathbf{x})\lambda_1 - p^{-1}\lambda_1 = 0 \\ &\iff \lambda_1 = (\|L_G V(\mathbf{x})\|^2 + p^{-1})^{-1} F_V(\mathbf{x})\end{aligned} \tag{A.29}$$

Replacing into the input equation A.27 the KKT multiplier formula A.29:

$$\begin{aligned}k_1(\mathbf{x}) &= -L_G V^\top(\mathbf{x})\lambda_1 \\ &= -L_G V^\top(\mathbf{x})(\|L_G V(\mathbf{x})\|^2 + p^{-1})^{-1} F_V(\mathbf{x})\end{aligned} \tag{A.30}$$

Syntethizing like this the controller $k(\mathbf{x})$ that complies all the necessary conditions:

$$\mathbf{k}(\mathbf{x}) = \begin{cases} \mathbf{k}_1(\mathbf{x}), & (\mathbf{x}) \in \mathcal{S}_1 \\ \mathbf{k}_2(\mathbf{x}), & (\mathbf{x}) \in \mathcal{S}_2 \\ \mathbf{k}_3(\mathbf{x}), & (\mathbf{x}) \in \mathcal{S}_3 \\ \mathbf{0}_{m\times 1}\ , & (\mathbf{x}) \in \mathcal{S}_4 \end{cases} \tag{A.31}$$

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